7.1 Rules That Fail for Matrices

WORKSHEET

The question

STEP II 1993, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q6

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself For real numbers, why is it true that $ab = 0$ implies $a = 0$ or $b = 0$? Which property of numbers does the argument use, and does it hold for matrices?

STAGE 1 Can $\mathbf{AB} = \mathbf{0}$ but $\mathbf{BA} \neq \mathbf{0}$?

Ask yourself Think geometrically. You want $\mathbf{B}$ to send everything into a line that $\mathbf{A}$ then squashes to zero. But in the other order, $\mathbf{A}$ should not squash everything that $\mathbf{B}$ then kills.

STAGE 3 Non-zero solutions of $\mathbf{AX} = \mathbf{0}$

Ask yourself If $\det \mathbf{A} \neq 0$, what can you do to $\mathbf{AX} = \mathbf{0}$? If $\det \mathbf{A} = 0$, can you find a non-zero column vector $\mathbf{v}$ with $\mathbf{Av} = \mathbf{0}$?

STAGE 4 **How many square roots?**

Pause and try Find non-zero $\mathbf{A}$ and $\mathbf{B}$ for which $\mathbf{X}^2 + \mathbf{AX} + \mathbf{B} = \mathbf{0}$ has infinitely many solutions. (*Hint:* make the left-hand side a perfect square, $(\mathbf{X} - \mathbf{I})^2$, and think about matrices whose square is zero.)

What's really going on

Matrices as an algebra

Arthur Cayley introduced matrix algebra in 1858, treating matrices as single objects that can be added and multiplied, rather than just as tables of coefficients. He noticed at once that multiplication is not commutative, and he proved the *Cayley–Hamilton theorem*: every square matrix satisfies its own characteristic equation. For a 2×2 matrix with trace t and determinant d, this says $\mathbf{M}^2 - t\mathbf{M} + d\mathbf{I} = \mathbf{0}$, a fact we use repeatedly in this chapter (Section 7.4). The 2×2 real matrices form a *ring*: you can add, subtract and multiply, but multiplication is not commutative, and there are *zero divisors*, non-zero elements whose product is zero.

Infinitely many square roots

Stage 4 can be made even more striking. The equation $\mathbf{X}^2 = \mathbf{I}$ has the obvious solutions $\pm\mathbf{I}$, but also every reflection of the plane in a line through the origin, $\mathbf{X} = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$, since reflecting twice does nothing. That is a whole circle of square roots of $\mathbf{I}$. A polynomial of degree n with number coefficients has at most n roots because numbers have no zero divisors; matrices do, and the count fails.

Non-commutativity in physics

In classical mechanics, position x and momentum p are numbers and $xp = px$. In quantum mechanics they become operators (infinite matrices) with $xp - px = \mathrm{i}\hbar$, where $\hbar$ is Planck's constant divided by 2π. This single non-commutation relation implies Heisenberg's uncertainty principle: position and momentum cannot both be known exactly. When Heisenberg discovered the rules of quantum mechanics in 1925 he did

not know what matrices were; Max Born recognised that the strange multiplication rule Heisenberg had invented was matrix multiplication.

Try this next

1 **When do they commute?** Find all 2×2 matrices that commute with $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

2 **Square roots of zero.** Find all 2×2 real matrices $\mathbf{N}$ with $\mathbf{N}^2 = \mathbf{0}$. (*Hint:* use the Cayley–Hamilton theorem to show $\operatorname{tr}\mathbf{N} = \det\mathbf{N} = 0$.)

3 **Reflections.** Show that every reflection matrix $\begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$ squares to $\mathbf{I}$, and find the line it reflects in.

4 **Cayley–Hamilton.** Prove that every 2×2 matrix satisfies $\mathbf{M}^2 - (\operatorname{tr}\mathbf{M})\mathbf{M} + (\det\mathbf{M})\mathbf{I} = \mathbf{0}$ by direct calculation.

Further reading

- **Gilbert Strang, *Introduction to Linear Algebra*** (Wellesley-Cambridge). The most popular first linear algebra text, with an emphasis on understanding.
- **Sheldon Axler, *Linear Algebra Done Right*** (Springer; free online). A more abstract approach, centred on linear maps rather than matrices.
- **Jim Baggott, *The Quantum Story*** (Oxford University Press). The history of quantum mechanics, including Heisenberg's matrix mechanics.

The full guided solution is in Section 7.1:
https://beyondstep.keithmcnulty.org/ch07/matrix-rules.html